

Social Relevant Project

Second Year

BASIC STUDENT DETAILS

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Domain : Software

BASIC PROJECT DETAILS

Project Title : Entrepreneurship Facilitation Services Processing Centre

Problem Bank Reference / ID : S01889

Domain : Software

Application Area : Entrepreneurship Support & MSME Services

SDG Number(s) & Title(s) : SDG 8 – Decent Work and Economic Growth. The project supports aspiring entrepreneurs and MSMEs by simplifying access to government schemes, registrations, and guidance, thereby promoting self-employment, startups, and sustainable economic growth.

PHASE 1: PROBLEM IDENTIFICATION

1.1 Problem Statement:

Introduction: Many students, young entrepreneurs, small business owners, and everyday citizens dream of starting their own business, but the journey often feels overwhelming. They don't know where to start, what registrations or licenses they need, or which government schemes or financial support they might be eligible for.

Problem: This challenge is especially common for first-time entrepreneurs and people living in semi-urban or rural areas. The processes can be confusing, requiring visits to multiple government offices or reliance on middlemen, which often discourages them from pursuing their ideas.

Solution: Entrepreneurship Facilitation Services Processing Centre is a digital platform that simplifies business registration, document processing, and access to support services. It ensures faster, transparent, and efficient service delivery for entrepreneurs.

1.2 Social Relevance:

☒ Health & Safety

☒ Time Loss

☒ Financial Loss

☐ Environmental Impact

☒ Quality of Life

Justification : Entrepreneurs often lose valuable time and money running from one office to another, unsure of the right steps to take. By simplifying access to guidance and support services, we can help them save resources, reduce frustration, and open the door to better income opportunities and a higher quality of life

1.3 Location & Context:

Location Type:

☒ Urban

☒ Semi-Urban

☐ Rural

Specific Location: Educational institutions, city transport routes, logistics operations

Scale of Impact:

☐ Individual

☒ Community

☒ Regional

1.4 Affected person details:

Problem facer Category	Level of impact	Importance to address this issue	Data available (if any)
Students	M	Safety & punctuality	Travel timing data
Young Entrepreneurs	H	Child safety	Bus route info
Small Business Owners	H	Route efficiency	Manual logs
Transport managers	H	Fleet monitoring	Vehicle records

1.5 Stakeholder Identification:

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Stakeholder Category	Domain mapping	Details	Any website identified
Students	Education	Daily commuters	NA
Transport Dept.	Public Services	Bus operations	NA
Logistics Companies	Transportation	Fleet owners	NA
Vehicle Owners	Personal Transport	Private users	NA
Developers	Software	System builders	NA
Network / GPS Service Providers	ICT Infrastructure	Provide internet connectivity	NA
Local Authorities	Public Administration	Use vehicle data for regulation, safety, and traffic management	NA
Logistics / Delivery Companies	Logistics & Supply Chain	Track delivery vehicles to reduce delays and improve efficiency	NA

PHASE 2: FIELD STUDY & BENEFICIARY SURVEY

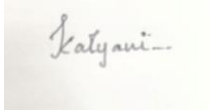
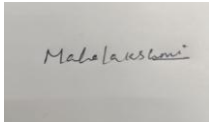
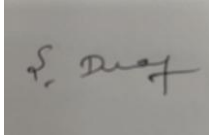
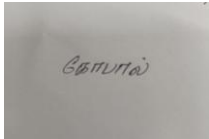
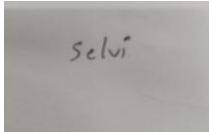
2.1 Purpose of Field Study:

- ☒ Understand real challenges
- ☒ Validate problem
- ☒ Study current practices
- ☒ Collect requirement data
- ☒ Identify Limitations

2.2 Field Visit Details:

Item	Details
Date of Visit	26 January 2026 – 28 January 2026
Time	Morning (7:30 AM – 9:00 AM) and Evening (4:30 PM – 6:00 PM)
Location	College bus stops, nearby main road junctions, hostel pickup points
Location Type	Urban and Semi-Urban
Purpose	To observe vehicle arrival delays, understand tracking difficulties, and interact with daily commuters
Participants Involved	Students, parents, bus drivers, delivery personnel
Outcome	Identified lack of real-time information, dependence on phone calls, and strong need for a simple tracking system

2.3 Beneficiary Profile :

ID	Age Group	Occupation	Location Type	Contact (Name, designation, organization, phone number)	Signature
A1	18-30	Employee	Semi urban	Varuna sri B. tech AIML 9629152690	
A2	31-45	Employee	Urban	Susi K, Agricultural shop Owner	
A3	40-50	Market owner	Semi urdan	Ravi Kumar, Proprietor, Local Retail Store 8883453696	
A4	40-50	Shop owner	rural	Karthi K, Small shop Owner	
A5	30-40	Shop owner	Semi urban	Mani K, Agricultural Owner	

2.4 Interaction Survey Questions

Beneficiary details: Person 1

A. Problem Occurrence & Impact

1. How often do you face this problem?

☐ Daily ☒ Weekly ☐ Monthly ☐ Rarely

2. At what time does the problem mostly occur?

☒ Morning ☐ Afternoon ☒ Evening ☐ Night ☐ Anytime

3. How long does the problem usually last?

☐ < 10 minutes ☒ 10–30 minutes ☐ 30–60 minutes ☐ > 1 hour

4. Severity of the problem

☐ Very High ☐ High ☐ Moderate ☒ Low

5. Primary impact of the problem (*Select one*)

☐ Safety risk

☐ Financial loss

☐ Health issue

☒ Time loss

☐ Inconvenience only

B. Current Practice & Limitations

6. Current method used to handle the problem

☒ Manual method

☐ Existing tool / system

☐ External help (person/service)

☐ No solution available

7. Effectiveness of the current method

☐ Very effective

☐ Moderately effective

☒ Slightly effective

☐ Not effective

8. Main difficulty in the current method

☐ High cost

- ☐ Time consuming
- ☐ Difficult to operate
- ☒ Not reliable
- ☐ Not always available

9. Frequency of failure in current method

- ☐ Very frequent
- ☒ Sometimes
- ☐ Rare
- ☐ Never

C. Technology & Infrastructure Constraints

10. Comfort level in using technology

- ☒ Very comfortable
- ☐ Can use with guidance
- ☐ Difficult to use
- ☐ Cannot use at all

11. Electricity availability at location

- ☐ Always available
- ☒ Mostly available
- ☐ Rarely available
- ☐ Not available

12. Internet availability

- ☒ Good
- ☐ Limited
- ☐ Poor
- ☐ Not available

13. Device availability

- ☐ Smartphone
- ☐ Basic mobile
- ☐ Computer
- ☐ No device

D. Cost Sensitivity & Acceptance

14. Acceptable cost level

- ☐ No cost
- ☐ Low cost
- ☒ Medium cost
- ☐ High cost acceptable

15. Preference for solution type

- ☒ Software based
- ☐ Hardware based
- ☐ Process / method based
- ☐ Combination

16. Willingness to adopt a new solution

- ☐ Yes, immediately
- ☐ Yes, after testing
- ☒ Maybe
- ☐ No

17. Willingness to recommend the solution to others

- ☒ Yes
 - ☐ No
-



Beneficiary Profile -Person 2

A. Problem Occurrence & Impact

6. How often do you face this problem?

☐ Daily ☒ Weekly ☐ Monthly ☐ Rarely

7. At what time does the problem mostly occur?

☒ Morning ☐ Afternoon ☒ Evening ☐ Night ☐ Anytime

8. How long does the problem usually last?

☐ < 10 minutes ☐ 10–30 minutes ☒ 30–60 minutes ☐ > 1 hour

9. Severity of the problem

☐ Very High ☐ High ☐ Moderate ☒ Low

10. Primary impact of the problem (Select one)

☐ Safety risk

☐ Financial loss

☐ Health issue

☒ Time loss

☐ Inconvenience only

B. Current Practice & Limitations

10. Current method used to handle the problem

☐ Manual method

☒ Existing tool / system

☐ External help (person/service)

☐ No solution available

11. Effectiveness of the current method

- ☐ Very effective
- ☒ Moderately effective
- ☐ Slightly effective
- ☐ Not effective

12. Main difficulty in the current method (*Select one*)

- ☐ High cost
- ☒ Time consuming
- ☐ Difficult to operate
- ☐ Not reliable
- ☐ Not always available

13. Frequency of failure in current method

- ☐ Very frequent
- ☒ Sometimes
- ☐ Rare
- ☐ Never

C. Technology & Infrastructure Constraints

14. Comfort level in using technology

- ☐ Very comfortable
- ☒ Can use with guidance
- ☐ Difficult to use
- ☐ Cannot use at all

15. Electricity availability at location

- ☐ Always available
- ☒ Mostly available
- ☐ Rarely available
- ☐ Not available

16. Internet availability

- ☐ Good
- ☒ Limited
- ☐ Poor

☐ Not available

17. Device availability

☒ Smartphone

☐ Basic mobile

☐ Computer

☐ No device

D. Cost Sensitivity & Acceptance

18. Acceptable cost level

☐ No cost

☐ Low cost

☒ Medium cost

☐ High cost acceptable

19. Preference for solution type

☒ Software based

☐ Hardware based

☐ Process / method based

☐ Combination

20. Willingness to adopt a new solution

☐ Yes, immediately

☐ Yes, after testing

☒ Maybe

☐ No

21. Willingness to recommend the solution to others

☒ Yes

☐ No



Beneficiary details: Person 3

A. Problem Occurrence & Impact

11. How often do you face this problem?

☐ Daily ☒ Weekly ☐ Monthly ☐ Rarely

12. At what time does the problem mostly occur?

☒ Morning ☐ Afternoon ☒ Evening ☐ Night ☐ Anytime

13. How long does the problem usually last?

☐ < 10 minutes ☒ 10–30 minutes ☐ 30–60 minutes ☐ > 1 hour

14. Severity of the problem

☐ Very High ☐ High ☐ Moderate ☒ Low

15. Primary impact of the problem (*Select one*)

☐ Safety risk

☐ Financial loss

☐ Health issue

☒ Time loss

☐ Inconvenience only

B. Current Practice & Limitations

14. Current method used to handle the problem

☒ Manual method

☐ Existing tool / system

☐ External help (person/service)

☐ No solution available

15. Effectiveness of the current method

☐ Very effective

☐ Moderately effective

☒ Slightly effective

☐ Not effective

16. Main difficulty in the current method (Select one)

☐ High cost

☐ Time consuming

☐ Difficult to operate

☒ Not reliable

☐ Not always available

17. Frequency of failure in current method

☐ Very frequent

☒ Sometimes

☐ Rare

☐ Never

C. Technology & Infrastructure Constraints

18. Comfort level in using technology

☒ Very comfortable

☐ Can use with guidance

☐ Difficult to use

☐ Cannot use at all

19. Electricity availability at location

☐ Always available

☒ Mostly available

☐ Rarely available

☐ Not available

20. Internet availability

☒ Good

☐ Limited

☐ Poor

☐ Not available

21. Device availability

- ☒ Smartphone
- ☐ Basic mobile
- ☐ Computer
- ☐ No device

D. Cost Sensitivity & Acceptance

22. Acceptable cost level

- ☐ No cost
- ☐ Low cost
- ☒ Medium cost
- ☐ High cost acceptable

23. Preference for solution type

- ☒ Software based
- ☐ Hardware based
- ☐ Process / method based
- ☐ Combination

24. Willingness to adopt a new solution

- ☐ Yes, immediately
- ☐ Yes, after testing
- ☒ Maybe
- ☐ No

25. Willingness to recommend the solution to others

- ☒ Yes
- ☐ No



Beneficiary details: Person 4

A. Problem Occurrence & Impact

16. How often do you face this problem?

☒ Daily ☐ Weekly ☐ Monthly ☐ Rarely

17. At what time does the problem mostly occur?

☒ Morning ☐ Afternoon ☒ Evening ☐ Night ☐ Anytime

18. How long does the problem usually last?

☐ < 10 minutes ☒ 10–30 minutes ☐ 30–60 minutes ☐ > 1 hour

19. Severity of the problem

☐ Very High ☐ High ☐ Moderate ☒ Low

20. Primary impact of the problem (*Select one*)

☐ Safety risk

☐ Financial loss

☐ Health issue

☒ Time loss

☐ Inconvenience only

B. Current Practice & Limitations

18. Current method used to handle the problem

☒ Manual method

☐ Existing tool / system

☐ External help (person/service)

☐ No solution available

19. Effectiveness of the current method

- ☐ Very effective
- ☐ Moderately effective
- ☒ Slightly effective
- ☐ Not effective

20. Main difficulty in the current method (*Select one*)

- ☐ High cost
- ☐ Time consuming
- ☐ Difficult to operate
- ☒ Not reliable
- ☐ Not always available

21. Frequency of failure in current method

- ☐ Very frequent
- ☒ Sometimes
- ☐ Rare
- ☐ Never

C. Technology & Infrastructure Constraints

22. Comfort level in using technology

- ☒ Very comfortable
- ☐ Can use with guidance
- ☐ Difficult to use
- ☐ Cannot use at all

23. Electricity availability at location

- ☐ Always available
- ☒ Mostly available
- ☐ Rarely available
- ☐ Not available

24. Internet availability

- ☒ Good
- ☐ Limited

- ☐ Poor
- ☐ Not available

25. Device availability

- ☐ Smartphone
- ☐ Basic mobile
- ☐ Computer
- ☐ No device

D. Cost Sensitivity & Acceptance

26. Acceptable cost level

- ☐ No cost
- ☐ Low cost
- ☒ Medium cost
- ☐ High cost acceptable

27. Preference for solution type

- ☒ Software based
- ☐ Hardware based
- ☐ Process / method based
- ☐ Combination

28. Willingness to adopt a new solution

- ☐ Yes, immediately
- ☐ Yes, after testing
- ☒ Maybe
- ☐ No

29. Willingness to recommend the solution to others

- ☒ Yes



Beneficiary details: Person 5

A. Problem Occurrence & Impact

21. How often do you face this problem?

☒ Daily ☐ Weekly ☐ Monthly ☐ Rarely

22. At what time does the problem mostly occur?

☒ Morning ☐ Afternoon ☒ Evening ☐ Night ☐ Anytime

23. How long does the problem usually last?

☐ < 10 minutes ☒ 10–30 minutes ☐ 30–60 minutes ☐ > 1 hour

24. Severity of the problem

☐ Very High ☐ High ☐ Moderate ☒ Low

25. Primary impact of the problem (*Select one*)

☐ Safety risk

☐ Financial loss

☐ Health issue

☒ Time loss

☐ Inconvenience only

B. Current Practice & Limitations

22. Current method used to handle the problem

☒ Manual method

☐ Existing tool / system

☐ External help (person/service)

☐ No solution available

23. Effectiveness of the current method

- ☐ Very effective
- ☐ Moderately effective
- ☒ Slightly effective
- ☐ Not effective

24. Main difficulty in the current method (*Select one*)

- ☐ High cost
- ☐ Time consuming
- ☐ Difficult to operate
- ☒ Not reliable
- ☐ Not always available

25. Frequency of failure in current method

- ☐ Very frequent
- ☒ Sometimes
- ☐ Rare
- ☐ Never

C. Technology & Infrastructure Constraints

26. Comfort level in using technology

- ☒ Very comfortable
- ☐ Can use with guidance
- ☐ Difficult to use
- ☐ Cannot use at all

27. Electricity availability at location

- ☐ Always available
- ☒ Mostly available
- ☐ Rarely available
- ☐ Not available

28. Internet availability

- ☒ Good
- ☐ Limited
- ☐ Poor

☐ Not available

29. Device availability

☐ Smartphone

☐ Basic mobile

☐ Computer

☐ No device

D. Cost Sensitivity & Acceptance

30. Acceptable cost level

☐ No cost

☐ Low cost

☒ Medium cost

☐ High cost acceptable

31. Preference for solution type

☒ Software based

☐ Hardware based

☐ Process / method based

☐ Combination

32. Willingness to adopt a new solution

☐ Yes, immediately

☐ Yes, after testing

☒ Maybe

☐ No

33. Willingness to recommend the solution to others

☒ Yes

☐ No

2.5 Field Observations

A. Operational Issues

☒ Workarounds observed

☒ Repeated manual effort

- ☒ Process delays
- ☒ Dependency on others

B. Safety & Risk Factors

- ☒ Safety hazards present
- ☐ Risk of injury
- ☐ Health-related risk
- ☐ No safety issues observed

C. Resource & Efficiency Issues

- ☒ Resource wastage (time/material/energy)
- ☐ High operational cost
- ☒ Inefficient workflow
- ☒ Redundant steps

D. User Behaviour Patterns

- ☐ Users ignore the problem
- ☒ Users adapt temporary fixes
- ☒ Users depend on external help



Observation Summary

During the field visit, frequent and unpredictable delays in vehicle arrival were observed at pickup and drop locations. Users were seen repeatedly contacting drivers, students, or staff to confirm vehicle location and

arrival time. No centralized or reliable real-time tracking system was available to provide accurate information. This led to time wastage, confusion, and increased dependency on others for updates.

PHASE 3: REQUIREMENT STUDY

3.1 Functional Requirements

Req ID	Requirement Description	Derived From (Survey Q.No / Observation)	Priority
FR-01	System shall show real-time vehicle location	Q.No 1,2	High
FR-02	System shall provide delay alerts	Q.No 3	High
FR-03	System shall display route history	Observation	Medium
FR-04	System shall allow multiple user access	Survey	Medium
FR-05	System shall work on mobile devices	Q-C	High

3.2 Non-Functional Requirements

A. Usability & Accessibility

- ☒ Easy to learn
- ☒ Minimal user input
- ☐ Local language support
- ☐ Suitable for low literacy users

B. Cost & Affordability

- ☐ No recurring cost
- ☒ Low initial cost
- ☐ Affordable for mass users

C. Performance & Reliability

- ☐ Fast response
- ☒ Accurate output
- ☐ Works consistently under normal conditions

D. Safety & Compliance

- ☐ Physically safe to use
- ☒ Data privacy protected
- ☐ Complies with legal norms

E. Maintenance & Sustainability

- ☒ Low maintenance
 - ☐ Easily repairable
 - ☐ Environment friendly
-

3.3 Constraints Analysis

A. Infrastructure Constraints

- ☒ Limited power supply
- ☐ Power outages
- ☐ Limited internet connectivity
- ☐ No internet availability

B. User-Related Constraints

- ☒ Low technical literacy
- ☐ Resistance to change
- ☐ Dependence on external help

C. Environmental Constraints

- ☒ Outdoor usage
- ☐ Dust / heat / humidity exposure
- ☒ Seasonal variations

D. Legal & Safety Constraints

- ☒ Safety regulations
 - ☐ Government policies
 - ☐ Ethical constraints
-

3.4 Mandatory vs Optional Requirements

Requirement ID	Requirement Description	Type	Reason
FR-01	Live tracking	Mandatory	Core functionality
FR-02	Alerts	Mandatory	Safety & time
FR-03	Route history	Optional	Improves usability
FR-04	Multi-user access	Optional	Enhances performance

- ☒ Requirements feasible under constraints
- ☒ Requirements aligned with social relevance
- ☒ Requirements suitable for next semester execution

PHASE 4: JOURNEY MAPPING

4.1 Objective of Journey Mapping

- ☒ Understand complete user experience
- ☒ Identify exact pain points
- ☒ Detect hidden inefficiencies
- ☒ Identify intervention opportunities
- ☒ Validate requirement assumptions

4.2 User Journey Map

Step No	User Action	User Thought	User Feeling	Problem Occurrence (Y/N)	Pain Point Description
1	Passenger waits for bus	When will bus arrive?	Frustrated	Y	No real-time tracking
2	Calls driver/transport office	Will they respond?	Irritated	Y	Communication delay
3	Driver starts vehicle	Hope route is clear	Neutral	N	Communication delay
4	Traffic congestion occurs	Running late	Stressed	Y	No live traffic update
5	Fleet manager checks records	Need vehicle status	Confused	Y	Manual record maintenance
6	Parent waits for school bus	Is child safe?	Anxious	Y	No live monitoring

4.3 Pain Point Identification

Major Pain Points Identified:

- 1.Lack of centralized and structured guidance
- 2.Heavy dependence on middlemen
- 3.Financial loss due to service charges
- 4.Poor awareness about government schemes
- 5.No proper tracking mechanism

Minor Pain Points Identified:

- 1.Language barriers
- 2.Confusion in documentation
- 3.Delay in communication from authorities

4.4 Opportunity Areas for Engineering Intervention

Pain Point	Root Cause	Possible Intervention Area
Communication delay	No integrated platform	Mobile app with notifications
Manual monitoring	No centralized dashboard	Cloud-based admin dashboard
Safety concern	No live alerts	Emergency alert system

4.5 Journey Map Summary

- 1.Most users do not know the correct starting point for business registration.
- 2.Due to confusion, they depend on agents and informal sources.
- 3.This results in financial loss and delays.
- 4.There is no single trusted platform providing complete support.
- 5.A structured digital facilitation system can significantly simplify the process

PHASE 5: BOUNDARY CONDITIONS & CONSTRAINT ANALYSIS

5.1 Boundary Conditions

Category	Boundary Condition Identified	Reason
Government Rules	Transport regulations	Legal compliance
Road Infrastructure	Existing roads	Cannot modify
Vehicle Type	Already purchased vehicles	Cost constraint
Budget	Limited funding	Student project

5.2 System Constraints

Infrastructure Constraints:

- ☒ Limited power supply
- ☒ Frequent power cuts
- ☐ Limited internet
- ☐ No internet

User Constraints:

- ☒ Low literacy
- ☐ Low digital skills
- ☐ Resistance to change

Environmental Constraints:

- ☒ Outdoor usage
- ☒ High temperature
- ☒ Dust / moisture exposure

Safety & Legal Constraints:

- ☒ Government approval required
- ☐ Safety compliance required
- ☒ Data privacy concern

5.3 Current System Bottlenecks

Bottleneck	Cause	Impact Level (H/M/L)
No real-time updates	Manual system	H
Data delay	Poor network	H
Tracking errors	Low GPS accuracy	M

5.4 Feasibility Risk Identification

The project may face risks such as internet connectivity issues affecting real-time tracking, high initial cost of GPS and GSM devices, and maintenance challenges. There may also be limitations in technical skills required for system implementation. However, these risks can be managed through proper planning, cost optimization, and reliable network support.

5.5 Constraint Analysis Summary

- Internet connectivity varies across regions.
- Users are cost-sensitive and require simple interfaces.
- Data privacy must be ensured at all stages.

PHASE 6: SOLUTION IDEATION

6.1 Solution Concept – 1

Concept Type:

- ☐ Software-Based
- ☐ Hardware-Based
- ☐ Process-Based
- ☒ Hybrid

Core Idea:

GPS + GSM-based tracking device integrated with mobile app.

How It Solves the Problem:

Provides real-time vehicle location and instant updates to users.

Advantages:

- Real-time monitoring
- Improves passenger safety

Limitations:

- Requires internet
- Hardware cost

Feasibility Level:

- ☒ High
- ☐ Moderate
- ☐ Low

6.2 Solution Concept – 2

Concept Type:

- ☒ Software-Based

☐ Hardware-Based

☐ Process-Based

☐ Hybrid

Core Idea:

Mobile application with driver update feature.

How It Solves the Problem:

Driver manually updates location and delay status.

Advantages:

- Low cost
- Easy to implement
- Depends on driver input
- Less accurate

Limitations:

Feasibility Level:

☒ High

☐ Moderate

☐ Low

6.3 Solution Concept – 3

Concept Type:

☐ Software-Based

☒ Hardware-Based

☐ Process-Based

☐ Hybrid

Core Idea:

RFID-based checkpoint tracking system.

How It Solves the Problem:

Vehicle is detected at fixed checkpoints.

Advantages:

- Works without internet
- Low maintenance

Limitations:

- Not continuous tracking
- Installation cost

Feasibility Level:

☐ High

☒ Moderate

☐ Low

6.4 Solution Concept – 4 Concept

Type:

☐ Software-Based

☐ Hardware-Based

☐ Process-Based

☒ Hybrid

Core Idea:

IoT-based smart tracking with cloud dashboard and AI route optimization.

How It Solves the Problem:

Uses analytics to optimize route and monitor performance.

Advantages:

- Advanced analytics
- Fuel efficiency

Limitations:

- Complex development
- Higher cost

Feasibility Level:

- ☐ High
- ☒ Moderate
- ☐ Low

6.5 Solution Concept – 5

Concept Type:

- ☐ Software-Based
- ☐ Hardware-Based
- ☒ Process-Based
- ☐ Hybrid

Core Idea:

Centralized control room with manual monitoring.

How It Solves the Problem:

Operators track and coordinate vehicles.

Advantages:

- Low technical complexity
- Human supervision
- Labor intensive

Limitations:

- Less scalable

Feasibility Level:

- ☐ High
- ☒ Moderate
- ☐ Low

6.6 Comparative Analysis of All Concepts

Criteria	Concept 1	Concept 2	Concept 3	Concept 4	Concept 5
Initial Cost	Medium	Low	Medium	High	Medium
Maintenance Cost	Medium	Low	Low	High	High (Manpower)
Real-Time Tracking	Yes (Continuous)	Partial	No (Only at checkpoints)	Yes (Advanced real-time)	No
Accuracy Level	High (5–10m GPS accuracy)	Low	Moderate	Very High	Low
Scalability	High	Moderate	Low	Very High	Low

Ideation Summary

1. Five different solution concepts were developed.
2. Hybrid GPS-based tracking provides best balance of cost and performance.
3. Software-only solution lacks accuracy.
4. AI-based system is powerful but costly.
5. Concept 1 is selected for further development.

Declaration:

- ☒ Journey mapping is based on real beneficiary interaction
- ☒ Constraints derived from field study
- ☒ Minimum 5 solution concepts developed
- ☒ No premature selection without comparison

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PHASE 7: BENEFICIARY VALIDATION OF SOLUTIONS

7.1 Objective of Beneficiary Validation

- ☒ Verify whether the proposed solutions actually address user needs
- ☒ Identify practical usability issues
- ☒ Understand beneficiary acceptance level
- ☒ Collect improvement suggestions from real users
- ☒ Avoid designing solutions that users will not adopt

7.2 Beneficiary Interaction Details

Interaction No	Beneficiary Type	Location	Mode (Interview / Discussion / Observation)	Date
1	Student (Bus User)	College Bus Stop	Interview	03-04-2026
2	Parent	Residential Area	Discussion	03-04-2026
3	Bus Driver	Transport Parking Area	Interview	03-03-2026
4	Office Employee	City Bus Stop	Observation	03-04-2026
5	Transport Manager	College Transport Office	Discussion	03-04-2026

7.3 Solution Concepts Presented to Beneficiaries

Concept No	Concept Description (Short)	Beneficiary Details	Beneficiary Response	Acceptance Level (High/Medium/Low)
Concept 1	GPS + GSM Real-Time Tracking System	Students, Drivers	Found very useful for tracking live location	High
Concept 2	Driver Manual Update Mobile App	Drivers	Easy but depends on manual updates	Medium
Concept 3	RFID Checkpoint Tracking	Manager	Limited tracking accuracy	Low
Concept 4	IoT Smart Tracking with AI	Manager	Powerful but costly	Medium
Concept 5	Manual Control Room	Manager	Labour intensive	Low

7.4 Beneficiary Feedback Summary Positive Feedback:

1. Real-time tracking will reduce waiting time.
2. Parents feel safer if they can track vehicles.
3. Notifications will help manage time efficiently.

Concerns Raised by Beneficiaries:

1. Internet connectivity may affect performance.
2. Hardware installation cost should be affordable.
3. Device maintenance must be simple.

Suggestions for Improvement:

1. Add delay notification alerts.
2. Provide easy-to-use mobile interface.
3. Include emergency alert feature.

7.5 Accepted vs Rejected Ideas

Concept	Accepted / Rejected	Reason
Concept 1	Accepted	Provides accurate real-time tracking
Concept 2	Accepted	Useful backup method
Concept 3	Rejected	Limited real-time tracking
Concept 4	Rejected	High cost and complexity
Concept 5	Rejected	Requires continuous manpower

7.6 Validation Summary

Beneficiaries supported the real-time vehicle tracking system for the Entrepreneurship Facilitation Processing Center as it improves safety and reduces waiting time. Most preferred GPS-based automatic tracking over manual methods. Some concerns included internet issues and device cost. Users suggested features like delay alerts and emergency notifications. Overall, Concept 1 received the highest acceptance and meets user needs effectively

PHASE 8: PATENT & EXISTING SOLUTION SEARCH

8.1 Objective

- ☒ Identify existing patents related to the problem
- ☒ Study existing market solutions
- ☒ Understand limitations of current approaches
- ☒ Ensure originality of proposed solution

8.2 Patent Search Summary

Patent Title	Patent Number	Country	Key Idea	Relevance to Project
Vehicle Tracking System Using GPS	US20130289945 A1	USA	GPS-based vehicle monitoring	Highly relevant
Real-Time Fleet Monitoring System	WO2014178943 A1	International	Real-time vehicle monitoring	Relevant
Smart Vehicle Tracking Device	IN20184102345 6	India	GSM-based tracking	Relevant

Sources Used:

- ☒ Google Patents
- ☒ WIPO
- ☒ Indian Patent Database

8.3 Existing Products / Solutions in Market

Product / System Name	Company / Organization	Key Features	Limitations	URL Link of the Product/System Name
GPS Tracker	LocoNav	Real-time tracking, alerts	Subscription cost	View LocoNav Vehicle Tracking System
Fleet Management System	TrackoBit	Route monitoring	Complex setup	https://trackobit.com
Vehicle GPS Device	MapmyIndia	Navigation & tracking	Expensive hardware	https://www.mapmyindia.com

8.4 Research Paper / Academic Work Review :

Several research studies highlight the effectiveness of GPS and GSM-based vehicle tracking systems in improving transportation efficiency and safety. These systems provide real-time location updates, reduce delays, and enhance monitoring capabilities. Existing academic work also emphasizes the importance of integrating alert systems for emergencies and delays. The review supports the feasibility and reliability of the proposed solution for practical implementation.

8.5 Identified Gaps in Existing Solutions

1. High cost of existing solutions
2. Complex interfaces for users
3. Limited customization for small-scale users

8.6 Justification for Proposed Solution

Explain why your proposed solution is still needed:

Existing solutions are often expensive and designed for large-scale fleet management, making them unsuitable for small institutions and individual users. Many systems require complex setup and high maintenance costs. The proposed solution aims to provide an affordable, simple, and scalable vehicle tracking system suitable for schools, colleges, and small transport services. It focuses on user-friendly design and efficient real-time tracking. Therefore, the proposed system fulfills the need for a low-cost and accessible tracking solution.

PHASE 9: FINAL CONCEPT SELECTION

9.1 Final Selected Concept Selected

Concept Name:

GPS + GSM Based Real-Time Vehicle Tracking System Concept

Type:

☒ Software

☒ Hybrid

9.2 Selection Criteria

Criteria	Concept 1	Concept 2	Concept 3	Concept 4	Concept 5
Beneficiary Acceptance	High	Medium	Low	Medium	Low
Technical Feasibility	High	High	Medium	Medium	Medium
Cost Feasibility	Medium	Low	Medium	High	Medium
Implementation Complexity	Medium	Low	Medium	High	Medium
Impact Potential	High	Medium	Low	High	Low

9.3 Reason for Final Selection

Concept 1 was selected because it provides continuous real-time vehicle tracking with high accuracy. It offers better safety and operational efficiency compared to other concepts. Beneficiaries showed strong acceptance of this solution. The system is technically feasible within available resources and budget. Therefore, it is the most suitable solution among all alternatives.

9.4 SDG Mapping of Final Concept

- SDG Number : 8
- SDG Title : **SDG 8** → supports entrepreneurship and business activities
- **Justification of Mapping** : The solution supports entrepreneurship by improving transport efficiency and reliability, enabling better access to services and business opportunities. It enhances productivity, reduces delays, and contributes to economic growth, aligning with SDG 8.
- Explanation of how the solution contributes to the SDG targets: The solution contributes to SDG 8 by supporting entrepreneurship through efficient and reliable transport services. It also promotes innovation using GPS and GSM technologies (SDG 9) and improves mobility, reduces waiting time, and enhances safety (SDG 11), contributing to sustainable development.

9.5 Expected Impact of the Solution

Impact Area	Expected Outcome
Social Impact	Improved passenger safety
Economic Impact	Reduced fuel wastage
Environmental Impact	Optimized route usage
Usability Impact	Easy vehicle tracking

PHASE 10: RESOURCE, TECHNOLOGY & EXPERTISE MAPPING

10.1 Hardware Requirements

Component	Quantity	Purpose
GPS Module	1	Location tracking
GPS Module	1	Data communication
Microcontroller	1	System control

10.2 Software / Technology Stack

Technology / Tool	Purpose
Android Studio	Mobile app development
Embedded C	Microcontroller programming
IoT Platform	Data monitoring

Examples:

- ☒ Mobile App Development
- ☒ Web Platform
- ☒ Embedded Systems
- ☐ IoT Sensors
- ☐ Machine Learning
- ☒ Data Analytics

10.3 Skills Required

Skill	Availability
Programming	Available
AI/ML	Need to Learn
UI /UX	Need to Learn

10.4 External Support Required

Support Type	Source
Faculty Guidance	College
Technical Support	Online
Lab Facility	College
Government / Community Support	MSME

10.5 Skill Gap Analysis

1. AI model building
2. API integration
3. UI/UX design

PHASE 11: TIMELINE & PROJECT PLAN

11.1 Implementation Plan for Next Semester

Phase	Activity	Duration	Expected Output
Phase 1	Detailed Design	2	System design
Phase 2	Prototype Development	4	Prototype
Phase 3	Testing	2	Tested model
Phase 4	Improvement & Iteration	2	Optimized system
Phase 5	Final Demonstration	1	Final product

11.2 Milestones

Design Completion	13/04/2026
Prototype Ready	11/05/2026
Testing Completed	25/05/2026
Final Evaluation	08/06/2026

11.3 Risk Factor

Risk	Probability	Impact	Mitigation Plan
Technical issues	M	H	Testing
Low user adoption	M	H	Simple UI
Time delay	M	M	Proper planning

11.4 Execution Readiness Summary

Final Declaration

- ☒ Beneficiary validation conducted for solution concepts
- ☒ Required resources and technologies identified
- ☒ Patent and existing solution search completed
- ☒ Final concept selected based on structured analysis
- ☒ Implementation plan prepared for next semester

Signature of the Student